

HALAH — CA2 Research Report

Required Edits to Align the Report with the Implemented System (v1.2, Sepolia)

Every passage marked in red below is new or corrected wording to be inserted into Yasir_CA2_Report, in the author's own style. 73 edits, reviewed by six independent reviewer/editor passes and verified against the deployed prototype.

How to apply this edit pack

Each edit below names the exact place in the report (page, section, table, figure, algorithm or equation number), quotes the current wording, explains in one or two sentences why it no longer matches the implemented system, and then gives the **replacement or insertion text in red**. The red text is written in the same narrative voice as the rest of the report, so it can be pasted directly. Edits are grouped chapter by chapter, in reading order. Five corrections are systemic — they recur throughout the document — and are stated once here so they can be applied everywhere consistently:

R1 — The consent tuple is six elements, everywhere.

Every occurrence of $C = (I, A, P, E)$ — including Eq. (7), Eq. (36) and Eq. (51) — must become:

$$C = (I, A, P, T, L, E)$$

where I is the patient identity, A the authorised requester, P the purpose/data scope, $T = [t_s, t_e]$ the temporal validity window, L the jurisdiction, and E the guardian (endorsement) set. The implemented data model, the smart contract ConsentSBTV3 and the audit ledger all store the six-element form.

R2 — The decision rule has five verdicts, everywhere.

Every occurrence of the three-term rule $ALLOW = VI \wedge VA \wedge VP$ must become:

$$ALLOW = VI \wedge VA \wedge VP \wedge VT \wedge VL$$

with VT the temporal check against $[t_s, t_e]$ and VL the jurisdiction check. Revocation is evaluated inside VA . This is exactly the five-row decision matrix the prototype prints for every access request.

R3 — VL is jurisdiction matching, not location tracking.

Wherever VL is described, define it as:

$$V_L(C, R) = 1 \Leftrightarrow Jur(R) = Jur(C) \vee Jur(C) = \emptyset$$

i.e., the requester's declared jurisdiction must equal the consent's jurisdiction unless the consent leaves it unrestricted. The prototype demonstrates this with the AU-consent / US-requester denial.

R4 — Guardian approval is N-of-N, with $N = |G|$.

The threshold is not a free parameter k of a k -of- n scheme in the implemented version: the state machine requires **all** designated guardians to produce a valid EIP-712 signature before minting, so $N = |G|$ (evaluated with $|G| = 2$). Any k -of- n wording must either be corrected to N-of-N or explicitly marked as future work.

R5 — Delivered work is written in the past/present tense.

Chapters 5–9 repeatedly describe as future ("will be implemented", "is planned") what the v1.2 prototype has now delivered: Sepolia deployment at 0x0EC3617581DEC63A7f2bb162DC14d088916ADc73, token #1 minted in block 11,771,243 and revoked on-chain in block 11,771,307, token #2 as an independent reproduction in block 11,771,581, a 60-event audit ledger, and an independent verifier (verify_poc) reporting ALL CHECKS PASSED. These passages must switch from "will" to "has been", with the evidence cited.

Overview of the 73 edits

Report part	Edits	Text	Tables	Equations	Algorithms	Figures
Chapter 1 - Introduction (incl. Table 1)	9	7	2	0	0	0
Chapter 2 - Systematic Literature Review	4	2	1	1	0	0
Chapters 3–4 - Research Questions and Objectives	11	8	1	2	0	0
Chapter 5 - Research Architecture (MVC)	12	4	2	4	0	2
Chapter 6 - Proposed Solution (6.1–6.3)	13	3	0	2	8	0
Chapter 6 - Proposed Solution (6.4–6.7)	14	9	0	3	2	0
Chapters 7–9 - Progress, Publications and Plan	10	7	2	1	0	0
Total	73	40	8	13	10	2

Chapter 1 - Introduction (incl. Table 1)

Edit 1 [TABLE] Page 8, Table 1, RO5 row

Current (as written in the report): *Next / RO5: Temporal and Lifecycle Trust*

Why it must change: Temporal/lifecycle trust is now implemented (VT enforcement, on-chain notBefore/expiry, end-to-end revocation, six-state lifecycle).

Replace with / insert (add this in RED in the report):

Stage: Current. Objective: Extend the authorisation model with validity periods, expiration, revocation, and controlled consent-state transitions. Implemented in the current prototype: temporal windows are enforced locally and mirrored on-chain, and revocation is executed as a verifiable Sepolia transaction.

Edit 2 [TABLE] Page 8, Table 1, RO6 row

Current (as written in the report): *Next / RO6: Formal Security Assurance and Empirical Validation*

Why it must change: The empirical half of RO6 is substantially delivered (23 automated tests, live jurisdiction denial on Sepolia, independent chain verification); only formal analysis remains future.

Replace with / insert (add this in RED in the report):

Stage: Current (empirical) / Next (formal). Objective: Empirically evaluate the HALAH implementation using representative, negative, and adversarial scenarios (delivered through automated testing and live on-chain evaluation), and subsequently define and formally analyse the security properties of the framework.

Edit 3 [TEXT] Page 7, §1.3

Current (as written in the report): *the framework will be expanded to investigate temporal validity and lifecycle trust, including the effects of validity periods, expiration, revocation, and controlled state transitions*

Why it must change: No longer future work; all four elements are implemented.

Replace with / insert (add this in RED in the report):

The framework has been extended to incorporate temporal validity and lifecycle trust. Validity periods and expiration are enforced at evaluation time and mirrored on-chain, revocation is executed as a verifiable blockchain transaction, and consent progresses through controlled state transitions, reflecting the principle that trust cannot be regarded as permanently established at the moment a consent is created.

Edit 4 [TEXT] Page 7, §1.3

Current (as written in the report): *This will be complemented by adversarial and negative testing, as well as comprehensive empirical evaluation using the HALAH prototype*

Why it must change: Adversarial/negative testing and prototype evaluation have now been performed.

Replace with / insert (add this in RED in the report):

Adversarial and negative testing has been undertaken using the HALAH prototype, including per-dimension denial scenarios, signature-rejection tests, and zero-knowledge tamper and expiry tests, together with a live negative evaluation on the Sepolia network. Formal security analysis remains the subject of the next research stage.

Edit 5 [TEXT] Page 7, §1.3

Current (as written in the report): *next stage will also investigate formal security assurance and comprehensive empirical validation.*

Why it must change: Lowercase sentence start; only the formal half is still future.

Replace with / insert (add this in RED in the report):

The next stage will investigate formal security assurance; empirical validation has now been carried out on the implemented prototype.

Edit 6 [TEXT] Page 7, §1.3 closing

Current (as written in the report): *current research is concerned primarily with establishing the conceptual and technical foundations*

Why it must change: Missing definite article and capital letter after the full stop.

Replace with / insert (add this in RED in the report):

The current research is concerned primarily with establishing the conceptual and technical foundations through which trustworthy multi-party authorisation can be constructed.

Edit 7 [TEXT] Page 7, §1.3 closing

Current (as written in the report): *Next stage will build upon these foundations by investigating how authorisation trust can be maintained dynamically over time, formally reasoned about ... and comprehensively validated through prototype-based experimentation.*

Why it must change: Dynamic maintenance and prototype experimentation are now implemented; only formal reasoning remains.

Replace with / insert (add this in RED in the report):

The implemented prototype now maintains authorisation trust dynamically over time and has been validated through prototype-based experimentation; the next stage will build upon these foundations by reasoning formally about the model under defined security assumptions.

Edit 8 [TEXT] Page 7, §1.3

Current (as written in the report): *The subsequent next stage research will extend this foundation towards dynamic trust management and stronger security assurance.*

Why it must change: Doubled “subsequent next stage”; the dynamic-trust half is delivered.

Replace with / insert (add this in RED in the report):

The research has since extended this foundation towards dynamic trust management; stronger formal security assurance remains for the subsequent stage.

Edit 9 [TEXT] Page 7, §1.3 (before Table 1)

Current (as written in the report): *The distinction between the current contribution and the subsequent research is summarised in Table 1.*

Why it must change: Current-versus-subsequent framing is outdated.

Replace with / insert (add this in RED in the report):

Table 1 summarises the research dimensions, indicating those realised in the implemented HALAH prototype and the formal-assurance work that remains for subsequent research.

Chapter 2 – Systematic Literature Review

Edit 10 [TABLE] Page 20, Table 2, Proposed Framework row

Current (as written in the report): *Requires further extension towards lifecycle trust, formal assurance, and empirical validation*

Why it must change: Lifecycle trust and empirical validation are now implemented; only formal assurance remains.

Replace with / insert (add this in RED in the report):

Lifecycle trust and empirical validation are realised in the implemented prototype; formal security assurance requires further extension.

Edit 11 [TEXT] Page 18, §2.3.6 final paragraph

Current (as written in the report): *This research establishes the conceptual foundation for such integration. Future work can therefore investigate how trusted execution...*

Why it must change: Understates the contribution — an implemented, evaluated prototype now exists (the TEE direction itself remains future).

Replace with / insert (add this in RED in the report):

This research establishes the conceptual foundation for such integration and demonstrates it through an implemented and empirically evaluated prototype. Future work can therefore investigate how trusted execution and remote attestation can strengthen the enforcement and verification of decentralised consent decisions.

Edit 12 [TEXT] Pages 18–19, §2.3.7 lead-in

Current (as written in the report): *A system may successfully execute valid authorisation scenarios without demonstrating that invalid identities, insufficient approvals, or policy violations cannot produce an unintended access decision.*

Why it must change: Stated as an open concern, but the prototype empirically tests these negative cases.

Replace with / insert (add this in RED in the report):

(Append) The present prototype addresses this concern empirically through adversarial and negative testing across each verification dimension; formal analysis remains necessary to demonstrate these properties under defined assumptions.

Edit 13 [EQUATION] Page 21 (end of Ch.2), Eq (6)

Current (as written in the report): $Decision(C,R) = ALLOW \iff VI(u) \wedge VA(C) \wedge VP(C,R)$
... The subsequent research will extend this model towards temporal and lifecycle trust...

Why it must change: The implemented rule is five-predicate; the 'will extend' framing is stale.

Replace with / insert (add this in RED in the report):

$Decision(C, R, t) = ALLOW \iff VI \wedge VA \wedge VP \wedge VT \wedge VL$. This is the rule enforced by the implemented Consent Engine: every access request is evaluated over the complete tuple, and the temporal and jurisdictional predicates are already operational rather than deferred to subsequent research.

Chapters 3-4 – Research Questions and Objectives

Edit 14 [EQUATION] Page 24, §3.3.1, Eq (7)

Current (as written in the report): $C = (I, A, P, E)$

Why it must change: The tuple omits T (temporal window) and L (jurisdiction); the implemented and demonstrated tuple has six components. Apply the same correction wherever the four-component form recurs (Eq 36 p.43, Eq 51 p.51).

Replace with / insert (add this in RED in the report):

$C = (I, A, P, T, L, E)$, where I represents the participating identities, A the authority and multi-party authorisation evidence, P the purpose and data-scope constraints, T the temporal validity window, L the jurisdictional constraint, and E the enforcement state recording whether the consent remains active or has been revoked.

Edit 15 [EQUATION] Page 25, §3.3.3, after Eqs (9)–(10)

Current (as written in the report): let N denote the required authorisation threshold ...
 $VA(C) = 1$ if $|Av| \geq N$

Why it must change: Generic N invites a k-of-n reading; the prototype fixes $N = |G|$.

Replace with / insert (add this in RED in the report):

(Append) In the implemented framework N is instantiated as $|G|$, the number of designated guardians, so that activation requires the approval of every listed guardian; configurable k-of-n thresholds are identified as future work.

Edit 16 [TEXT] Page 23, §3.2

Current (as written in the report): *Temporal and lifecycle trust, formal security assurance, and comprehensive empirical validation form the subsequent research scope and will be developed further in CA3.*

Why it must change: Temporal/lifecycle trust is implemented; empirical validation partially complete.

Replace with / insert (add this in RED in the report):

Temporal and lifecycle trust has been operationalised in the current prototype, and the empirical validation has been partially completed through automated and live negative testing; formal security assurance forms the subsequent research scope and will be developed in CA3.

Edit 17 [TEXT] Page 29, §4 introduction

Current (as written in the report): *...while the final two objectives extend and evaluate these mechanisms as part of the next research programme.*

Why it must change: RO5 implemented, RO6 partially.

Replace with / insert (add this in RED in the report):

...while the fifth objective is realised operationally in the current prototype and the sixth is addressed empirically now, with its formal component reserved for the next research programme.

Edit 18 [TEXT] Page 30, §4.3 RO3

Current (as written in the report): *The mechanism will define the required authorisation participants, verify their approval evidence, and determine whether the predefined threshold has been satisfied.*

Why it must change: 'Will' for delivered work; 'threshold' needs its N-of-N instantiation.

Replace with / insert (add this in RED in the report):

The implemented mechanism defines the designated guardians at consent creation, verifies each EIP-712 approval by signature recovery, and requires a valid signature from every listed guardian: the threshold equals [G], an N-of-N condition demonstrated as 2-of-2.

Edit 19 [TEXT] Page 30, §4.4 RO4

Current (as written in the report): *Accordingly, the mechanism will evaluate whether the requested purpose is consistent with the consent and whether the requested data scope is contained within the permitted scope.*

Why it must change: Delivered work in future tense.

Replace with / insert (add this in RED in the report):

Accordingly, the implemented mechanism evaluates whether the requested purpose is consistent with the consent and rejects any request whose data scope is not a non-empty subset of the authorised EMR modules.

Edit 20 [TEXT] Pages 30–31, §4.5 RO5

Current (as written in the report): *This objective constitutes a principal component of the next stage research programme. It will investigate how temporal and lifecycle conditions can be incorporated...*

Why it must change: RO5 is substantially realised.

Replace with / insert (add this in RED in the report):

This objective has been substantially realised in the current implementation. The prototype enforces the temporal validity window at every access decision (VT), records notBefore and expiry on-chain within the consent token, implements end-to-end revocation including an on-chain revoke transaction, and manages the consent through an explicit lifecycle state machine. The remaining work concerns hardening these controls rather than introducing them.

Edit 21 [TEXT] Page 31, §4.6 RO6

Current (as written in the report): *Although preliminary implementation and scenario-based testing may support the current stage findings, the complete formal analysis and comprehensive empirical validation will be conducted as part of next stage works.*

Why it must change: Understates delivered evidence.

Replace with / insert (add this in RED in the report):

The empirical component has been partially delivered: 23 automated tests exercise per-dimension denial, signature rejection and proof tampering; a live out-of-jurisdiction request was refused on the deployed Sepolia contract; and an independent tool re-verified every recorded transaction against the chain. The complete formal analysis — proofs or model checking of the stated properties — remains next stage work.

Edit 22 [TEXT] Page 31, §4.7

Current (as written in the report): *RO5 and RO6 constitute the principal research agenda for CA3... The current stage research consequently concentrates on RO1-RO4...*

Why it must change: Contradicts the implementation status.

Replace with / insert (add this in RED in the report):

The current stage delivers RO1-RO5 in operational form and the empirical component of RO6, including negative and adversarial testing. The principal remaining agenda for CA3 is the formal component of RO6 — formal specification and analysis of the security properties — together with the identified prototype hardening.

Edit 23 [TEXT] Page 31, §4.7 (typo)

Current (as written in the report): summarised in Table 3 and Table 3.

Why it must change: Duplicate reference.

Replace with / insert (add this in RED in the report):
summarised in Table 3 and Table 4.

Edit 24 [TABLE] Page 33, Table 4, RO5/RO6 rows

Current (as written in the report): RO5 “Extend the authorisation model with temporal validity, expiration, revocation...” (listed under CA3)

Why it must change: RO5 is implemented; RO6 must distinguish done empirical work from future formal work.

Replace with / insert (add this in RED in the report):
RO5 scope: Temporal validity, expiration, revocation and controlled state transitions, enforced in the current prototype and anchored on-chain. RO6 scope: Formal analysis of the security properties (CA3), building on the empirical negative and adversarial testing already performed with the HALAH prototype.

Chapter 5 – Research Architecture (MVC)

Edit 25 [EQUATION] Pages 34, 42, 45 — Eqs (29), (35), (42), (43)

Current (as written in the report): $Decision(C,R) = ALLOW \iff V_I(u) \wedge V_A(C) \wedge V_P(C,R)$ — presented as the current-stage decision, with V_{TVL} deferred.

Why it must change: The implemented engine enforces all five checks and the Doctor view displays the five-check matrix.

Replace with / insert (add this in RED in the report):
Make $ALLOW = VI \wedge VA \wedge VP \wedge VT \wedge VL$ the decision rule throughout the chapter; delete each three-predicate variant or relabel it explicitly as an early design iteration superseded by the implementation.

Edit 26 [EQUATION] Page 34 Eq (30), Page 37 §5.1.2 Eq (31), Figure 5

Current (as written in the report): $V_L(C)$... determines whether the consent remains in a valid lifecycle state.

Why it must change: Symbol clash: in the implementation V_L is the jurisdiction check; lifecycle/revocation is enforced inside V_A .

Replace with / insert (add this in RED in the report):
 $V_L(C, j)$ verifies that the requester’s declared jurisdiction matches the consent’s jurisdiction L ; revocation and lifecycle state are enforced within V_A . Relabel Figure 5’s $V_L(C)$ box as “ $V_L(C, j)$ — Jurisdiction”.

Edit 27 [EQUATION] Page 43, Eq (36)

Current (as written in the report): $C = (I, A, P, E)$

Why it must change: Four-component tuple again.

Replace with / insert (add this in RED in the report):

$C = (I, A, P, T, L, E)$, with T (temporal window) and L (jurisdiction) defined as implemented.

Edit 28 [FIGURE / DIAGRAM] Page 36, Figures 3-4 (layered MVC)

Current (as written in the report): *Controller = {Request Validation, DID/VC Verification, Authority Verification, Multi-Party Threshold Evaluation, Policy & Scope Evaluation, Authorisation Decision, Audit Event Generation}; Model = seven local state boxes.*

Why it must change: No blockchain, ZK, revocation or verifier appears anywhere in the diagrams.

Replace with / insert (add this in RED in the report):

FIGURE EDIT — (a) Controller layer: add “ZK Proof Service (Groth16 generation + verification)” feeding the Authorisation Decision, and “On-chain Anchor Service (asynchronous mint/revoke)”; (b) Model layer: add “ConsentSBTV3 (Sepolia)” beside a box relabelled “SQLite Audit Ledger”; (c) add a Revocation arrow from the Patient interface through the Controller to both the ledger and the contract; (d) add an external box “verify_poc (independent verifier)” reading both the ledger and Sepolia.

Edit 29 [FIGURE / DIAGRAM] Page 41, Figure 6 (end-to-end workflow)

Current (as written in the report): *Flow ends “Authorization Decision → ALLOW/DENY → Record Audit Evidence”.*

Why it must change: Missing anchoring, VT/VL checks, ZK step, revocation and verification.

Replace with / insert (add this in RED in the report):

FIGURE EDIT — after “Threshold Satisfied? Yes” insert “Mint SBT in local ledger” → “Background job anchors mint on Sepolia (async)”; insert V_T and V_L checks before the Authorisation Decision; add “Generate & verify Groth16 proof” on the ALLOW branch; after “Record Audit Evidence” append “Patient revokes → on-chain revoke tx → Auditor review + independent verify_poc”. Rename “Threshold Satisfied?” to “All | G| guardians signed? (N-of-N)”.

Edit 30 [TEXT] Pages 36-37, §5.1.1 Controller decomposition

Current (as written in the report): *...six principal functions...*

Why it must change: Omits two implemented services.

Replace with / insert (add this in RED in the report):

...eight principal functions..., adding “Cryptographic Proof Generation and Verification (Groth16)” and “On-chain Anchoring and State Synchronisation (Sepolia mint/revoke, background)”. Update Figure 4’s Controller row accordingly.

Edit 31 [TEXT] Page 44, §5.3 closing

Current (as written in the report): *The subsequent next research will extend this representation to include temporal validity, expiration, revocation, lifecycle transitions...*

Why it must change: All implemented.

Replace with / insert (add this in RED in the report):

The implementation realises this representation, including temporal validity, expiration, on-chain revocation and an explicit lifecycle state machine; formal security properties remain future work.

Edit 32 [TEXT] Page 47, §5.8

Current (as written in the report): *Blockchain or other tamper-resistant storage mechanisms may be integrated as infrastructure...*

Why it must change: Hedged as hypothetical; anchoring is implemented.

Replace with / insert (add this in RED in the report):

In the implementation, consent activation and revocation are anchored on the Sepolia blockchain via the ConsentSBTV3 soulbound contract; anchoring is asynchronous, and the ledger records each transaction hash and block number.

Edit 33 [EQUATION] Pages 40/42/43, Eqs (33), (37)

Current (as written in the report): $|A_v| \geq N$ (generic threshold N)

Why it must change: Implies configurable k-of-n.

Replace with / insert (add this in RED in the report):

(Append after Eq 37) In the implemented prototype $N = |G|$: every listed guardian must sign (2-of-2 in the evaluation); configurable k-of-n thresholds are future work.

Edit 34 [TEXT] Pages 36 & 40 — “DID / VC Verification”, “Verifiable Credentials (VCs)”

Current (as written in the report): (diagram/box labels and prose)

Why it must change: Overstated: VC verification is presently a DID-format check; the requester is not authenticated.

Replace with / insert (add this in RED in the report):

(Add limitation sentence in §5.2/§5.5) In the prototype, credential verification is a DID well-formedness check and the requester's role and jurisdiction are self-declared; wallet-held keys and authenticated credentials are future work.

Edit 35 [TABLE] Page 49, Table 5 (security properties)

Current (as written in the report): *Six properties; "Authorisation evidence ... future verification".*

Why it must change: Missing three demonstrated properties; independent verification exists.

Replace with / insert (add this in RED in the report):

TABLE EDIT — add rows: Temporal validity (VT enforced; on-chain notBefore/expiry); Jurisdictional constraint (VL; live SA-vs-AU denial); Revocability (patient revoke; on-chain revoke; fail-closed VA). Change "future verification" to "independent verification (verify_poc re-checks every transaction against Sepolia: ALL CHECKS PASSED)".

Edit 36 [TABLE] Page 50, Table 6, RO5/RO6 rows

Current (as written in the report): *RO5/RO6 phrased as expected contributions.*

Why it must change: RO5 is implemented; RO6 empirical work done, formal not.

Replace with / insert (add this in RED in the report):

TABLE EDIT — RO5 contribution: "Implemented: VT enforcement, lifecycle state machine, end-to-end revocation including on-chain". RO6 mechanism: "Empirical validation (automated tests, live negative test, independent chain verification); formal proofs remain future work".

Chapter 6 - Proposed Solution (6.1-6.3)

Edit 37 [EQUATION] Page 51, Eq (51)

Current (as written in the report): $C = (I, A, P, E)$

Why it must change: Six-component tuple implemented; this chapter contradicts the Implementation chapter.

Replace with / insert (add this in RED in the report):

$C = (I, A, P, T, L, E)$, where T represents the temporal validity window of the consent and L represents the jurisdiction in which the consent may be exercised.

Edit 38 [ALGORITHM] Page 55, Algorithm 1

Current (as written in the report): *Line 15 constructs the four-tuple; no initial state after Persist C.*

Why it must change: Missing T and L; missing PENDING_SIGNATURE state; nothing is anchored at creation.

Replace with / insert (add this in RED in the report):

Line 15: "Construct C = (I, A, P, T, L, E)". Insert after line 16 (Persist C): "state $\leftarrow$ PENDING_SIGNATURE $\triangleright$ nothing is anchored on-chain until all guardians approve".

Edit 39 [ALGORITHM] *Pages 53–54, Procedure 6.1.2 and closing sequence*

Current (as written in the report): *No step validates the jurisdiction L; sequence "... S_C $\rightarrow$ T $\rightarrow$ A $\rightarrow$ id_C $\rightarrow$ C $\rightarrow$ E" omits L.*

Why it must change: L is collected, stored on-chain and enforced (the negative demonstration is a VL failure).

Replace with / insert (add this in RED in the report):

Insert "Step 6a — Validate the Jurisdiction. The declared jurisdiction L is validated as a well-formed ISO alpha-2 territory code (AU in the demonstrated transactions) and bound to the consent object, since the access decision is later evaluated against it." Sequence: I_p $\rightarrow$ D $\rightarrow$ G $\rightarrow$ P_u $\rightarrow$ S_C $\rightarrow$ T $\rightarrow$ L $\rightarrow$ A $\rightarrow$ id_C $\rightarrow$ C $\rightarrow$ E.

Edit 40 [TEXT] *Page 53, Steps 1–2*

Current (as written in the report): *"checked against the available identity and credential verification mechanisms"; the doctor's "presented credentials".*

Why it must change: Overclaims: DID-format checks only; the doctor presents no credentials.

Replace with / insert (add this in RED in the report):

In the current prototype this validation is a structural check that each identifier is a well-formed DID; the doctor is not independently authenticated, and credential-based verification is deferred to the production design.

Edit 41 [EQUATION] *Pages 58–60, §6.2 Eq "VC_valid = ..." and Algorithm 2 lines 9–16*

Current (as written in the report): $VC_valid = Sig_VC \wedge Issuer_valid \wedge Subject_match \wedge Time_valid$ (presented as implemented).

Why it must change: In the running system VC verification is a DID-format check only.

Replace with / insert (add this in RED in the report):

This condition defines the target verification model. The present prototype implements V_I(u) as a structural DID-format check and records the outcome; cryptographic VC validation against the four conditions above is specified here as the production behaviour and is not yet exercised by the demonstration.

Edit 42 [TEXT] Page 56, §6.2.1

Current (as written in the report): *the Guardian workflow ... invokes VC verification before adding the guardian's signature*

Why it must change: The operative mechanism is the EIP-712 signature, never mentioned in §6.2.

Replace with / insert (add this in RED in the report):

In the HALAH implementation, the guardian's authority is evidenced by an EIP-712 typed-data signature bound to the consent, the guardian DID and the Sepolia chain identifier, verified by off-chain signer recovery; the VC check preceding it is presently a DID-format check, with the signing keys derived deterministically from the guardian DIDs in place of wallet-held keys.

Edit 43 [TEXT] Page 61, §6.3 around Eq (54)

Current (as written in the report): *The threshold mechanism establishes collective authorisation without requiring every possible participant to approve.*

Why it must change: False for the implementation: N-of-N.

Replace with / insert (add this in RED in the report):

In the current implementation the threshold is fixed to the full guardian set, $N = |G|$, so every designated guardian must approve before activation; the demonstrated configuration is therefore 2-of-2. A configurable k-of-n threshold is a direct generalisation left to future work.

Edit 44 [ALGORITHM] Page 63, Procedure 6.3.2, Step 2

Current (as written in the report): *The system obtains the threshold value $N \dots 1 \leq N \leq |G|$.*

Why it must change: N is not configurable; it is derived.

Replace with / insert (add this in RED in the report):

Step 2 — Derive the Authorisation Threshold. The system derives the threshold from the guardian set as $N = |G|$, so that activation requires a valid approval from every designated guardian.

Edit 45 [ALGORITHM] Page 66, Algorithm 3, Require line

Current (as written in the report): *Require: Consent object C , participant approval a , threshold N*

Why it must change: N is not an input.

Replace with / insert (add this in RED in the report):

Require: Consent object C, participant approval a — and insert after line 1: “N ← |G|”.

Edit 46 [ALGORITHM] Page 66, Algorithm 3, lines 2–3

Current (as written in the report): *Verify identity and authority of participant submitting a*

Why it must change: The actual verification is off-chain EIP-712 recovery.

Replace with / insert (add this in RED in the report):

2: signer ← RecoverEIP712(domain, {id_C, g, APPROVE}, sig_a) ▷ off-chain recovery.
3: if signer ≠ Addr(g), or the signature binds a different consent, guardian, or chain, then return DENY.

Edit 47 [ALGORITHM] Page 66, Algorithm 3, line 15

Current (as written in the report): *Record valid approval a*

Why it must change: Evidence kept is unspecified.

Replace with / insert (add this in RED in the report):

15: record signature_data[g] ← (signer, sig_a).

Edit 48 [ALGORITHM] Page 66, Algorithm 3, lines 18–19

Current (as written in the report): *Activate authorisation / Generate activation or token evidence*

Why it must change: Understates the concrete on-chain mechanism.

Replace with / insert (add this in RED in the report):

19: state ← PENDING_TOKEN. 20: submit mintConsentV3(addr_p, addr_r, keccak(P_u), t_s, t_e, L, burnAuth) via the owner account (background job; reconciled on next lookup). 21: on confirmation: state ← ACTIVE; record the token identifier.

Edit 49 [ALGORITHM] Page 66, Algorithm 3, terminal line (minor)

Current (as written in the report): *return ALLOW*

Why it must change: Collides with the access-decision ALLOW.

Replace with / insert (add this in RED in the report):

return ACTIVATED – avoiding conflation of consent activation with the access decision.

Chapter 6 – Proposed Solution (6.4–6.7)

Edit 50 [EQUATION] Page 67, Eq (59)

Current (as written in the report): $Decision(C,R) = ALLOW \iff V_I(u) \wedge V_A(C) \wedge V_P(C,R)$

Why it must change: The implemented decision is the five-way conjunction, logged per request.

Replace with / insert (add this in RED in the report):

$Decision(C, R, t) = ALLOW \iff V_I \wedge V_A \wedge V_P \wedge V_T \wedge V_L$, evaluated at every access request; the decision matrix is recorded in the audit ledger with each GRANTED or DENIED result.

Edit 51 [EQUATION] Pages 76–77, Eqs (64)–(65) and §6.4.2

Current (as written in the report): $V_L(C) = ValidState(C) \wedge ValidTransition(C)$

Why it must change: V_L is jurisdiction in the implementation; lifecycle is inside V_A ; jurisdiction is otherwise absent from 6.4–6.6.

Replace with / insert (add this in RED in the report):

$V_L(C, R) = 1 \iff Jur(R) = Jur(C) \vee Jur(C) = \emptyset$, where the jurisdiction is a two-letter ISO code stored on-chain; lifecycle state and revocation are evaluated within V_A . This check produced the demonstrated denial for jurisdiction SA against a consent issued for AU.

Edit 52 [ALGORITHM] Page 72, Algorithm 4, lines 21–26

Current (as written in the report): *if $S_R \notin S_C$ then ... end if → 26: Generate or verify access evidence*

Why it must change: Missing temporal, jurisdiction and non-empty-scope checks.

Replace with / insert (add this in RED in the report):

Line 21: “if $S_R = \emptyset$ or $S_R \notin S_C$ then ...”. Insert 25a: “if $t < t_s$ or $t > t_e$ then Record denied decision; return DENY”. Insert 25b: “if $Jur(R) \neq Jur(C)$ then Record denied decision; return DENY”.

Edit 53 [ALGORITHM] Page 72 Algorithm 4 line 26; Page 75 Step 8

Current (as written in the report): *Generate or verify access evidence*

Why it must change: The evidence is a concrete Groth16 proof, never named.

Replace with / insert (add this in RED in the report):

26: Generate and verify a Groth16 proof that $Poseidon(secret, expiry)$ equals the public commitment and that $t \leq t_e$; record $proof_mode = groth16$. Because the

prover and verifier currently run on the same server, the proof demonstrates toolchain integration rather than an additional privacy guarantee.

Edit 54 [TEXT] Page 74, Step 4

Current (as written in the report): *The valid approval set A_v is evaluated against the required threshold N*

Why it must change: Generic threshold again.

Replace with / insert (add this in RED in the report):

In the implementation the threshold is N-of-N: $N = |G|$, every designated guardian must sign (2-of-2 in the evaluated configuration); a configurable k-of-n threshold remains future work.

Edit 55 [TEXT] Page 74, Step 5

Current (as written in the report): *There is no execution path where the system can violate the privacy policy.*

Why it must change: Unproven formal claim.

Replace with / insert (add this in RED in the report):

In the prototype, purpose compatibility is an exact match against the consent purpose (stored on-chain as `keccak256(purpose)`); a formal no-violation proof is left to future work.

Edit 56 [TEXT] Page 71

Current (as written in the report): *the lifecycle validation mechanism can therefore be interpreted as a continuous validity layer.*

Why it must change: Validation is per-request, not continuous.

Replace with / insert (add this in RED in the report):

a per-request validity layer: validity is re-established at each evaluated access request and recorded as a `STATE_CHECK` event, rather than assumed between requests.

Edit 57 [TEXT] Page 69, Step 3

Current (as written in the report): *Typical lifecycle states may include `CREATED`, `PENDING`, `ACTIVE`, `SUSPENDED`, `REVOKED`, and `EXPIRED`.*

Why it must change: Does not match the implemented machine.

Replace with / insert (add this in RED in the report):

The implemented lifecycle states are `PENDING_SIGNATURE`, `PENDING_TOKEN`, `NOT_STARTED`, `ACTIVE`, `EXPIRED` and `REVOKED`.

Edit 58 [TEXT] Pages 76–77, §6.5.1

Current (as written in the report): *the complete temporal and lifecycle enforcement mechanism remains part of the planned CA3 extension.*

Why it must change: False — implemented.

Replace with / insert (add this in RED in the report):

Temporal and lifecycle enforcement is implemented: the Consent Engine evaluates $t_s \leq t \leq t_e$ at every request, the contract stores notBefore and expiry and exposes checkValid and checkValidAt, and revocation is executed both locally and as an on-chain revoke transaction (block 11,771,307).

Edit 59 [TEXT] Page 79, top (Algorithm 5 framing)

Current (as written in the report): *The algorithm is intended for the CA3 implementation and evaluation stage. It will be used to investigate expiration, revocation, and controlled state transitions...*

Why it must change: Algorithm 5 runs now, per request.

Replace with / insert (add this in RED in the report):

Algorithm 5 is implemented in the current prototype: its checks are executed at every access request and recorded as STATE_CHECK events in the audit ledger, so expiration, revocation and state transitions already condition the demonstrated access decisions.

Edit 60 [TEXT] Page 79, §6.6.1

Current (as written in the report): *The empirical component will evaluate both valid and invalid scenarios... Negative scenarios will examine whether access is denied...*

Why it must change: Executed already.

Replace with / insert (add this in RED in the report):

The empirical component has been executed against the prototype: 23 automated tests cover per-dimension matrix failures, EIP-712 signature rejection and zero-knowledge tamper and expiry cases; the live demonstration includes a jurisdiction-based denial and a revocation followed by refusal; and an independent verification tool re-checks every ledger transaction against Sepolia receipts, reporting ALL CHECKS PASSED.

Edit 61 [EQUATION] Page 79, §6.6, Eq (67)

Current (as written in the report): *For current stage, the primary validation conditions are $Decision(C,R) = ALLOW \Rightarrow V_I \wedge V_A \wedge V_P$.*

Why it must change: The tested property covers all five dimensions.

Replace with / insert (add this in RED in the report):

The validated soundness property is $Decision = ALLOW \Rightarrow V_I \wedge V_A \wedge V_P \wedge V_T \wedge V_L$, exercised by per-dimension failure tests for each conjunct.

Edit 62 [TEXT] Pages 79–80, §6.6 opening and Figure 12 caption

Current (as written in the report): *Formal security assurance... Empirical validation evaluates whether the implemented system behaves... (RO6 planned workflow)*

Why it must change: Conflates the status of the two halves.

Replace with / insert (add this in RED in the report):

The empirical half of this objective has been carried out on the prototype; formal proofs and model checking of the authorisation properties remain future work. Caption: “RO6 workflow for security assurance; the empirical branch has been executed, the formal branch remains planned.”

Edit 63 [TEXT] Page 86, §6.7

Current (as written in the report): *Algorithm 5 introduces temporal and lifecycle validity, while Algorithm 6 provides formal and empirical assurance... current stage establishes the foundational authorisation mechanisms and CA3 investigates their temporal extension.*

Why it must change: Out-of-date staging.

Replace with / insert (add this in RED in the report):

Algorithms 1–5 are all realised in the current prototype, and the empirical branch of Algorithm 6 has been executed against it; only the formal-assurance branch, together with k-of-n thresholds and authenticated requesters, remains for the next stage.

Chapters 7–9 – Progress, Publications and Plan

Edit 64 [EQUATION] Page 87, Eqs (69)–(70) framing

Current (as written in the report): *For the current stage implementation, the operational decision is based on Equation (69) [three terms], with Equation (70) presented as the CA3 extension.*

Why it must change: The implemented system enforces the full five-term rule of Eq. (70).

Replace with / insert (add this in RED in the report):

The implemented system enforces the complete rule of Equation (70): access is granted only when V_I , V_A , V_P , V_T and V_L all hold, with revocation evaluated within V_A , and every decision — granted or denied — is logged with the full matrix. Equation (69) is retained only as the historical current-stage formulation.

Current (as written in the report): *threshold-based multi-party authorisation ... required approval threshold N; RO2 identity verification*

Why it must change: Overstates generality; mechanism unstated.

Replace with / insert (add this in RED in the report):

(Append to §7.3) In the current prototype the threshold is fixed at N-of-N: every listed guardian must approve, each through an EIP-712 typed-data signature verified by cryptographic recovery, with the two-guardian case evaluated end-to-end. Configurable k-of-n thresholds remain future work. (Append to §7.2) In the prototype, VI performs identifier- and credential-format validation; cryptographic DID resolution against a registry and requester authentication are acknowledged limitations reserved for the next stage.

Current (as written in the report): *Status: In Progress ... The expected outcome is a policy-aware access-control mechanism... This objective will provide the final foundational component.*

Why it must change: Delivered.

Replace with / insert (add this in RED in the report):

Status: Completed at the prototype level. The policy predicate VP is implemented: each access request declares a purpose and a requested data scope S_R, and the request is rejected whenever S_R is not a subset of the authorised scope S_C. The predicate is evaluated as one term of the complete decision matrix $VI \wedge VA \wedge VP \wedge VT \wedge VL$, and every denial names the failed checks explicitly in both the interface and the audit ledger. Input validation on consent creation and access requests prevents malformed identifiers, scopes and jurisdictions from reaching the decision logic. Policy-aware access control is therefore an enforced property of the running system rather than an expected outcome.

Current (as written in the report): *Status: In Progress ... RO5 is reserved primarily for next stage ... A future consent validity interval will be represented as $[t_s, t_e]$.*

Why it must change: The most stale section in the report — RO5 is largely implemented.

Replace with / insert (add this in RED in the report):

Status: Largely Completed. RO5 has moved from planned work to implemented capability. The consent validity interval $[t_s, t_e]$ is now enforced by the temporal predicate VT at every access request and is additionally stored on-chain in the SBT as the notBefore and expiry fields, so temporal validity can be verified independently of the application. The consent lifecycle is implemented as an explicit state machine (PENDING_SIGNATURE, PENDING_TOKEN, NOT_STARTED, ACTIVE, EXPIRED, REVOKED), and revocation is enforced end-to-end: the patient's withdrawal updates

the application state and is confirmed as a revoke transaction on Sepolia in block 11,771,307, after which the contract itself reports the token as invalid and subsequent access requests are denied. What remains for the next stage is the dynamic and delegated dimension of lifecycle trust, such as consent modification, delegation and guardian-set changes, rather than temporal enforcement itself.

Edit 68 [TEXT] Pages 89–90, §7.6 (RO6) incl. Eq (80)

Current (as written in the report): *The current validation focuses primarily on RQ1, RQ2, and RQ3 ... Decision(C,R) = ALLOW iff VI $\wedge$ VA $\wedge$ VP.*

Why it must change: Understates validated coverage; Eq (80) omits VT and VL.

Replace with / insert (add this in RED in the report):

Status: Partially Completed. Empirical validation now covers all six trust dimensions. A suite of 23 automated tests exercises per-dimension failures of the decision matrix, rejection of invalid EIP-712 guardian signatures, and tampered or expired zero-knowledge proofs. Live validation on Sepolia includes a negative demonstration in which a request declaring jurisdiction SA against a consent issued for AU is denied with VL named as the failed check, and a revocation scenario in which the same consent, once revoked on-chain, is subsequently denied through VA. An independent verification tool re-checks every ledger transaction hash against Sepolia receipts and reads each token back from the contract, reporting that all checks passed, and every granted access is accompanied by a real Groth16 proof generated and verified at request time. The implemented decision rule is $\text{Decision}(C, R, t) = \text{ALLOW}$ iff $VI \wedge VA \wedge VP \wedge VT \wedge VL$. Formal security analysis and a systematic performance study remain for the next stage.

Edit 69 [TEXT] Page 90, §7.6 closing

Current (as written in the report): *The remaining ... activities will include policy-violation scenarios, temporal expiration, revocation, invalid lifecycle transitions, formal security properties, and broader experimental validation.*

Why it must change: Four of six items demonstrated.

Replace with / insert (add this in RED in the report):

Policy-violation scenarios, temporal expiration, revocation and invalid lifecycle transitions have now been demonstrated empirically, both in the automated test suite and in the live Sepolia deployment. The remaining work for the next stage is therefore narrower: formal definition and verification of the security properties, a systematic performance and overhead study, and the removal of prototype limitations such as N-of-N approval, server-derived guardian keys and the unauthenticated requester role.

Edit 70 [TABLE] Page 90, §8, Table 7

Current (as written in the report): *HALAH Prototype & Validation ... In Progress ... performance evaluation*

Why it must change: Implementation and initial validation complete on a public testnet.

Replace with / insert (add this in RED in the report):

TABLE EDIT — status: “Decentralised implementation and scenario-based validation complete on the Sepolia public testnet; performance evaluation outstanding.” The paper’s empirical section can now draw on the deployed contract, the two minted tokens and the verified transaction trail as delivered evidence.

Edit 71 [TEXT] Page 91, §9.1

Current (as written in the report): *RO1, RO2, and RO3 have been implemented ... RO4 represents the current transition towards policy-aware access control.*

Why it must change: Stale; omits the delivered Sepolia deployment.

Replace with / insert (add this in RED in the report):

RO1 to RO4 have been implemented and validated in the HALAH prototype, and RO5 is largely complete. The delivered evidence includes the ConsentSBTV3 contract deployed and source-verified on Sepolia at 0x0EC3617581DEC63A7f2bb162DC14d088916ADc73, with the evaluated consent minted as token #1 in block 11,771,243, revoked on-chain in block 11,771,307, and an independently reproduced lifecycle minted as token #2 in block 11,771,581. Each transaction hash is recorded in the audit ledger and publicly inspectable on Etherscan.

Edit 72 [TEXT] Page 91, §9.2

Current (as written in the report): *The first major focus of next stage is the introduction of temporal and lifecycle trust ... Next stage will therefore extend the consent model.*

Why it must change: Delivered focus listed as planned.

Replace with / insert (add this in RED in the report):

Temporal and lifecycle trust, originally planned for the next stage, has been brought forward and implemented: validity windows are enforced and anchored on-chain, revocation is effective at the contract level, and lifecycle transitions are governed by an explicit state machine. The next stage accordingly concentrates on two remaining foci: formal security assurance and comprehensive empirical validation, including performance measurement.

Edit 73 [TABLE] Page 92, Table 8

Current (as written in the report): *Row “Temporal and Lifecycle Trust” listed as planned next-stage activity.*

Why it must change: Delivered; empirical row partially delivered.

Replace with / insert (add this in RED in the report):

TABLE EDIT — mark Temporal and Lifecycle Trust as Completed (evidence: on-chain notBefore/expiry; revoke transaction in block 11,771,307; state machine); annotate

Comprehensive Empirical Validation as partially delivered (23 automated tests; live jurisdiction denial and revocation-then-denial on Sepolia; independent chain verification passed), with performance and overhead measurement remaining.

Closing note

Applying these 73 edits makes the CA2 report a faithful description of the system that now exists: the six-element consent tuple and five-verdict decision rule as implemented, the N-of-N EIP-712 guardian ceremony, the deployed **ConsentSBTv3** contract on Sepolia with a complete mint-and-revoke lifecycle that any third party can verify on Etherscan, and an audit trail whose every anchored event has been independently re-checked against chain receipts. No claim in the red text goes beyond what the prototype demonstrably does; conversely, everything the prototype still does not do (multi-jurisdiction deployment beyond the AU/SA pair, k-of-n thresholds, prover/verifier separation for the zero-knowledge component, and DID-registry key resolution) remains explicitly labelled as future work.